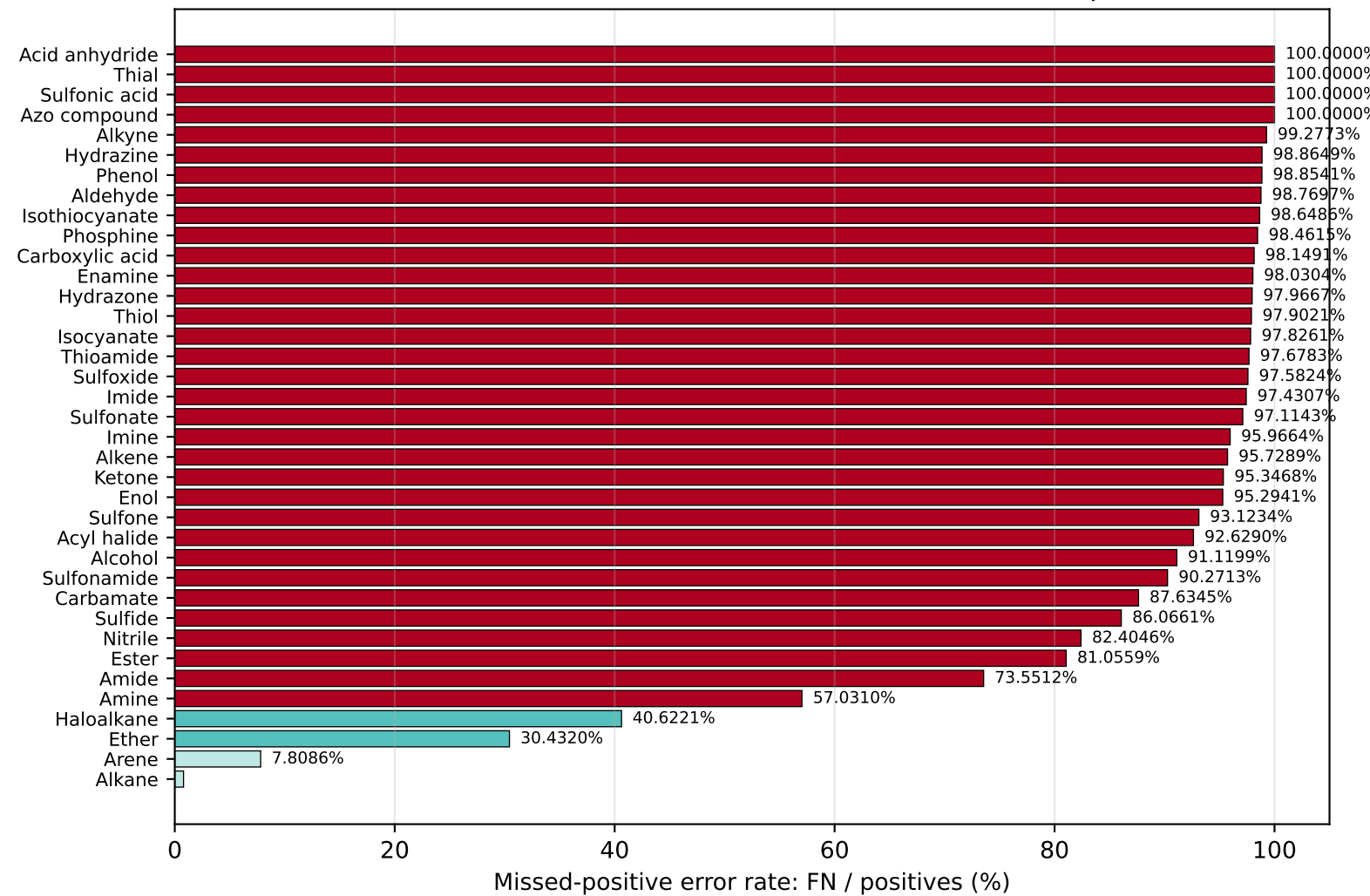
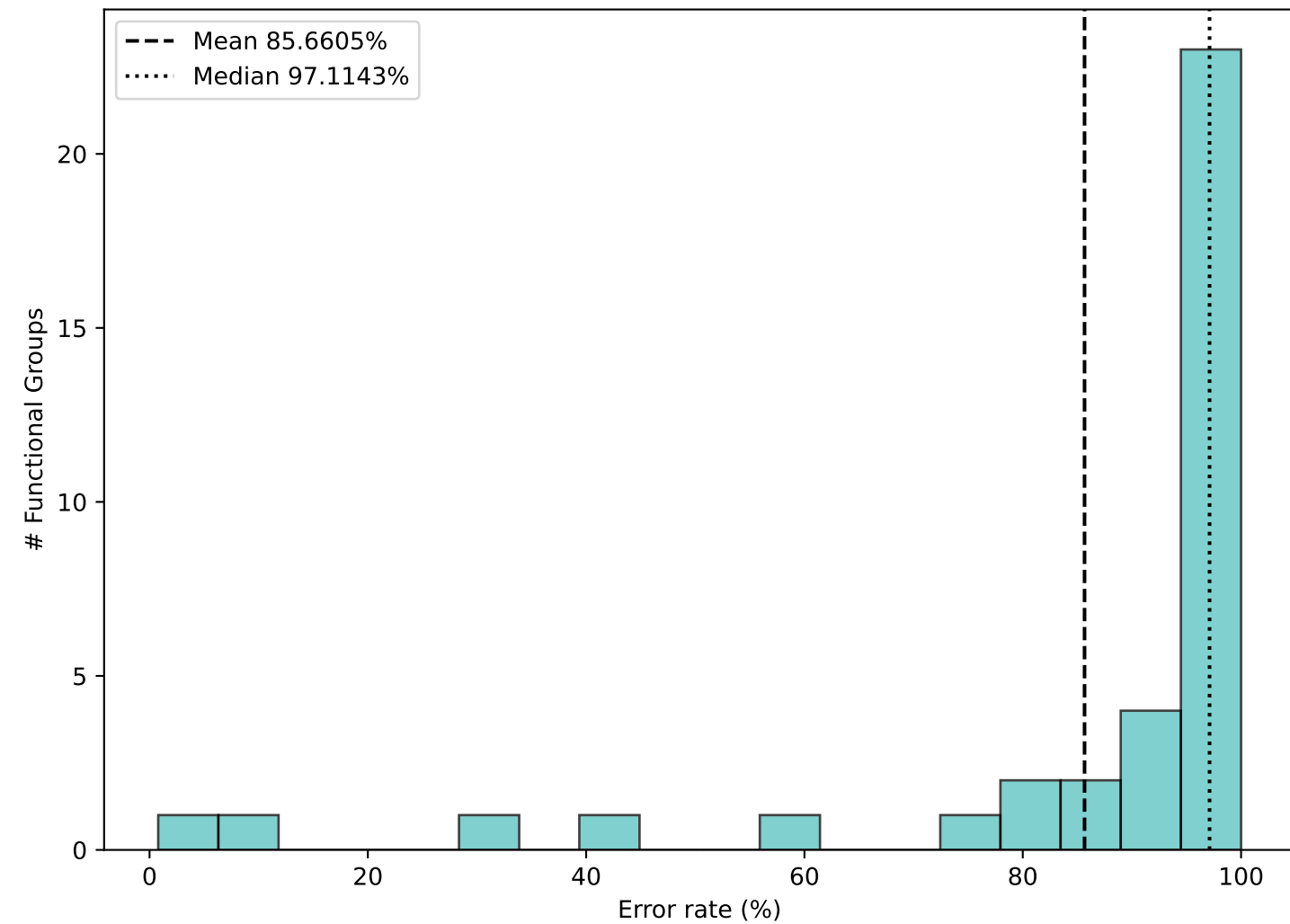


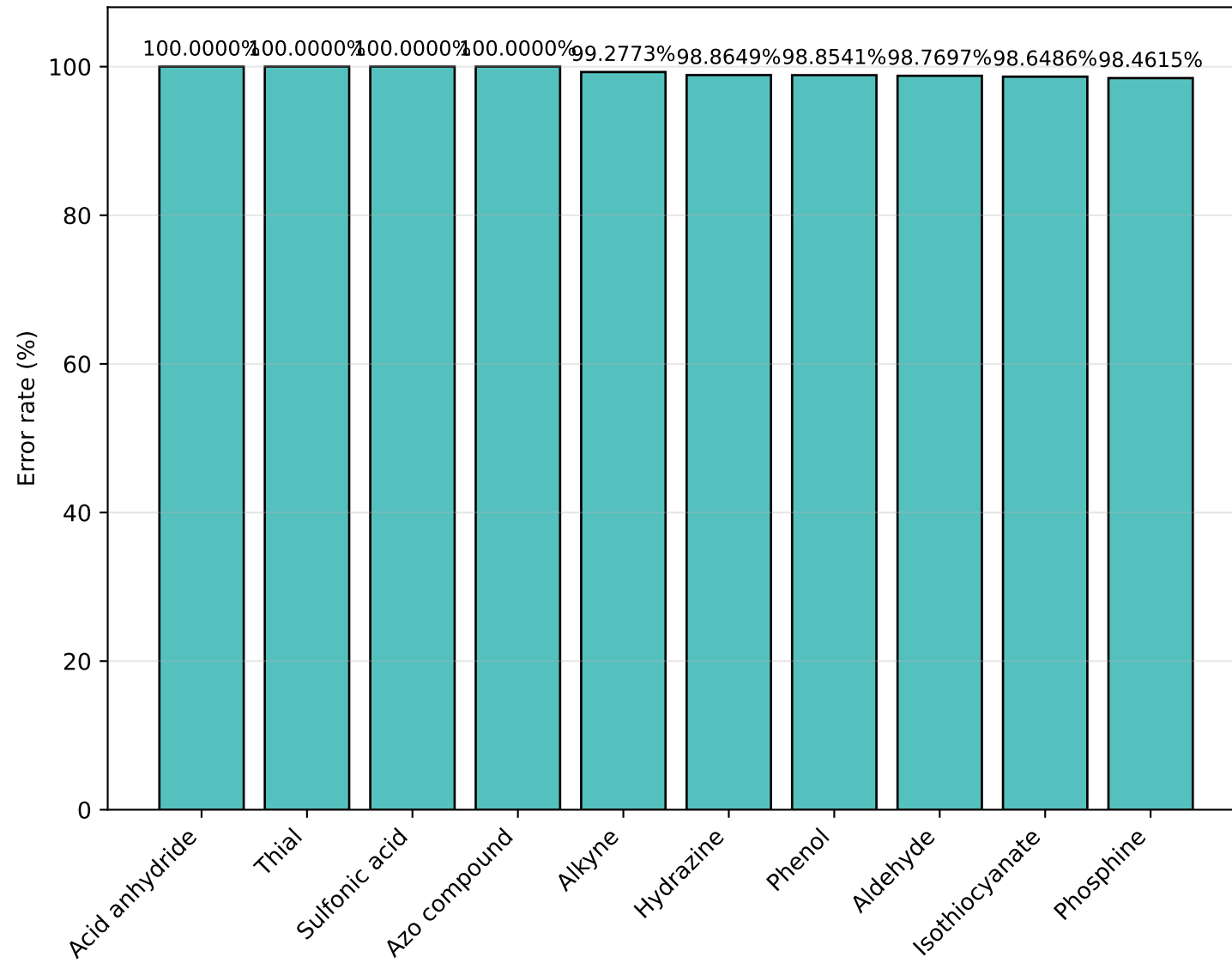
TTN MSMS+ Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



TTN MSMS+ VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.6306
F1 macro:	0.1576
Precision micro:	0.7101
Recall micro:	0.5672
Hamming accuracy:	0.9142

Error-rate statistics:

Mean:	85.6605%
Median:	97.1143%
Std:	25.3138%
Max:	100.0000% (Acid anhydride)
Min:	0.7972% (Alkane)

Problem classes:

Severe (>50%):	33 groups
High (20-50%):	2 groups
Moderate (5-20%):	1 groups
Low (<=5%):	1 groups